Experiment:24

Design a C program to demonstrate UNIX system calls for filemanagement.

Code:-

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/stat.h>
int main()
{
   int fd;
   char buffer[100];
   fd = creat("sample.txt", S_IRWXU);

   if (fd == -1)
   {
      perror("create");
      exit(1);
   }
   else
   {
      printf("File 'sample.txt' created successfully.\n");
      close(fd);
   }
   fd = open("sample.txt", O_WRONLY | O_APPEND);
```

```
if (fd == -1)
{
   perror("open");
   exit(1);
}
else
{
   printf("File 'sample.txt' opened for writing.\n");
}
write(fd, "Hello, World!\n", 14);
printf("Data written to 'sample.txt'.\n");

close(fd);
fd = open("sample.txt", O_RDONLY);

if (fd == -1)
{
   perror("open");
   exit(1);
}
else
{
   printf("File 'sample.txt' opened for reading.\n");
}
int bytesRead = read(fd, buffer, sizeof(buffer));
```

```
    if (bytesRead == -1)
    {
        perror("read");
        exit(1);
    }
    else
    {
        printf("Data read from 'sample.txt':\n");
        write(STDOUT_FILENO, buffer, bytesRead);
    }

    close(fd);
    if (remove("sample.txt") == -1)
    {
        perror("remove");
        exit(1);
    }
    else
    {
        printf("File 'sample.txt' deleted.\n");
    }

    return 0;
}
```

Output:

```
File 'sample.txt' created successfully.
File 'sample.txt' opened for writing.
Data written to 'sample.txt'.
File 'sample.txt' opened for reading.
Data read from 'sample.txt':
Hello, World!
File 'sample.txt' deleted.


...Program finished with exit code 0
Press ENTER to exit console.
```